

Audio Player using Tiva C LaunchPad and DAC

Group-03

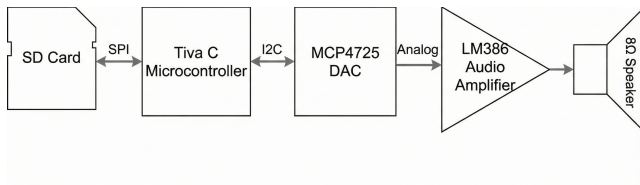
Embedded Systems Design

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- Real-time embedded audio playback is used in alarms, IoT devices, UI feedback, and portable systems.
- This project implements a standalone WAV audio player using:
 - Tiva C LaunchPad
 - FAT32 MicroSD card
 - MCP4725 12-bit I2C DAC
 - LM386 amplifier
 - 8ohm speaker
- No external DSP or codec IC — the entire pipeline runs on the MCU.

Overall System Architecture



- **SD Card → Tiva C → MCP4725 DAC → LM386 → 8ohm Speaker**
- Tiva handles file operations, decoding, buffering, timing, and DAC feed.
- LM386 amplifies the DAC signal to drive a real speaker.

Complete System Workflow Summary

- MCU boots → Configures clock → Initializes peripherals.
- Opens WAV file → Parses header → Locates PCM data.
- Prefills buf0 buf1 → Starts timer sample engine.
- ISR extracts samples → Feeds FIFO.
- Main loop writes samples → DAC → LM386 → Speaker.
- Playback ends when: both buffers empty + FIFO drained.

1. Power-Up and MCU Initialization

- Tiva configures PLL and runs at **80 MHz**.
- Enables all required peripherals:
 - GPIO (debug pin, speaker control)
 - I²C (DAC communication)
 - SPI (SD card interface)
 - Timer0A (audio sample clock)
- System is now electrically ready but audio hasn't started yet.

2. Loading Audio From SD Card

- SD card FAT32 filesystem is mounted.
- The MCU opens `speech.wav`.
- Reads first 12 bytes to confirm:
 - “RIFF” header
 - “WAVE” signature
- Walks through WAV chunks to extract:
 - Sample rate (8k)
 - Channels (must be 1)
 - Bits per sample (must be 16)
 - PCM data region

3. Pre-Filling Playback Buffers

- Two buffers are used:

buf0 (4 KB) buf1 (4 KB)

- Why double buffering?
 - buf0 → consumed by ISR
 - buf1 → refilled from SD
 - Ensures **zero gaps** during playback
- Both buffers are filled once before playback starts.

4. Starting the Audio Clock

- Timer0A is configured to tick exactly at the WAV file's sample rate.
- Example: 8 kHz PCM $\rightarrow$

$$T = 125 \mu s$$

- Every timer tick means:

“Play next sample now”

- The ISR provides perfect sample timing.

5. Timer ISR – Real-Time Sample Extraction

- ISR never touches I²C or SD card → **never blocks**.
- ISR operations:
 - Select active buffer (buf0 or buf1)
 - Extract one 16-bit PCM sample
 - Convert signed 16-bit → unsigned 12-bit for DAC
 - Push sample into FIFO
- Acts like a conductor ensuring timing accuracy.

6. FIFO — Shock Absorber Between ISR and DAC

- MCP4725 I²C write time varies.
- Timer ISR fires every 125 μ s.
- FIFO decouples these two timings.

FIFO Role

- ISR $\rightarrow$ Push samples at fixed rate
- Main loop $\rightarrow$ Pull samples at variable rate
- Prevents underrun/overflow

7. Main Loop – DAC Writer + Buffer Manager

A) DAC Writer

- Pops sample from FIFO
- Sends to MCP4725 via robust 3-byte I²C write
- Includes:
 - Error checking
 - Automatic 3 retries
 - Microsecond backoff

B) Buffer Manager

- Refill whichever buffer is not in use by ISR:

buf0 playing → refill buf1

buf1 playing → refill buf0

- Supports SD hiccups using retry-enabled reads.

Final Output: DAC $\rightarrow$ LM386 $\rightarrow$ 8Ω Speaker

- MCP4725 generates the analog audio waveform (0–3.3 V).
- LM386 amplifies the signal to 0.5–1W.
- Directly drives an speaker with clear audible output.

Hardware Components

- **Tiva C LaunchPad TM4C123**

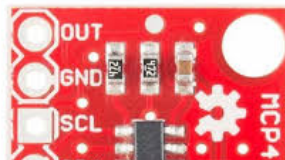
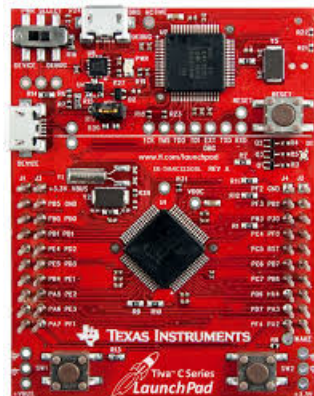
- ARM Cortex-M4F @ 80 MHz
- SPI (MicroSD interface)
- I2C (MCP4725 DAC communication)
- On-board debugging GPIO interfaces

- **MCP4725 – 12-bit DAC Module**

- I2C @ 400 kHz (Fast Mode)
- 0–3.3 V analog output
- 12-bit resolution (4096 steps)

- **8ohm Speaker**

- Driven through LM386 amplifier
- Suitable for 0.5–1 W audio output



Hardware Components

- **LM386 Audio Amplifier**

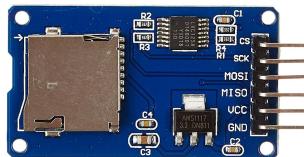
- Boosts DAC output for speaker use
- Operates on 5–12 V single supply

- **8 Speaker**

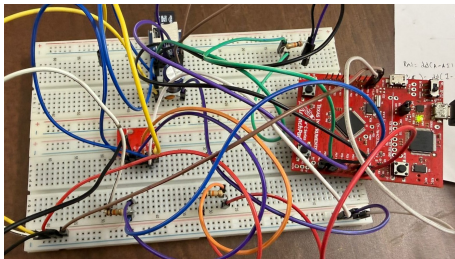
- Output power: 0.5–1 W using LM386
- Produces clear audible output from DAC signal

- **MicroSD Card Module**

- SPI-based SD card interface
- Supports FAT32 filesystems



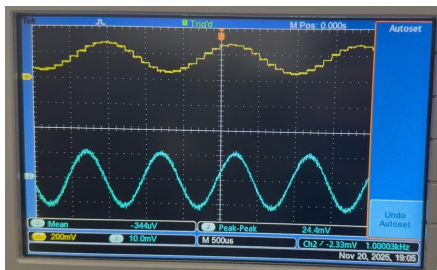
Real Circuit Implementation



- Demonstrates physical wiring of SD module, DAC module, LM386 amplifier and speaker.
- All signals verified on oscilloscope for stability.

Experimental Results

- Smooth playback at 8 kHz and 16 kHz.
- Verified analog output waveforms on oscilloscope.
- LM386 output successfully drives 8ohm speaker.
- FIFO effectively prevents underruns during SD delays.



Channel 1: DAC O/P
Channel 2: Original Signal

Conclusion

- Designed and implemented a fully standalone WAV audio player.
- Robust pipeline:
 - SD → Double Buffer → ISR → FIFO → DAC → LM386 → Speaker
- Reliable real-time playback with no codec IC.
- Extendable to:
 - Stereo audio
 - Volume control
 - Wireless/Bluetooth streaming

Thank You!